

# Optimising Collective Intelligence: A Comparative Analysis and Enhancement of RCTA Aggregation Methodologies

**Executive Summary:** This analysis evaluates five standard aggregation methods for the RCTA dataset and identifies the Geometric Mean of Odds as the most stable baseline. This suggests RCTA forecasters systematically under-reported their true confidence and benefited from having their predictions pushed toward more decisive outcomes. To improve further on this, I developed a Skill-Weighted Geometric Mean of Odds, which gives greater influence to forecasters with a proven track record of accuracy. This improved model reduced the error rate (Brier score) from 0.1104 to 0.1098, providing a sharper and more reliable signal for decision-making.

## Step 1

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To evaluate the RCTA competition's performance, I assessed three primary datasets that track the lifecycle of crowd wisdom from initial hunch to outcome:

- The ground truth ([rct-a-questions-answers.tab](#)): This serves as the definitive reference for 955 forecasting problems. It contains the specific event criteria, active windows, and the final resolved outcomes used to grade accuracy.
- The ledger of judgment ([rct-a-prediction-sets.csv](#)): A granular log of every update made by forecasters. This captures the raw logic of the crowd as they reacted to real-world news in real-time.
- The daily consensus ([rct-a-daily-forecasts.csv](#)): The largest dataset, providing a daily snapshot of the standing forecast for every participant. This allows us to analyse the collective consensus as a continuous time series.

For the final analysis, I focused on the two most robust sources to ensure computational efficiency and statistical consistency, the ground truth and daily consensus. While I also examined the [rct-a-prediction-sets.csv](#), which captures individual updates in real-time, I prioritised the daily consensus for the final benchmarks. This ensured models were built on the standing wisdom of the crowd, providing a more stable and representative signal. By linking these files via unique identifiers ([discover question id](#) and [discover answer id](#)), we can track how accuracy evolved day by day. This scale of

data (totalling over 1.4 GB) ensures that our aggregation benchmarks are mathematically robust and representative of the entire tournament's diversity.

## Step 2

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To ensure the accuracy of our benchmarks, I implemented a live-window filter that excludes administrative data points captured before a question is launched or after it has concluded. The analysis focuses exclusively on resolved problems, ensuring that our performance metrics (Brier scores) are grounded in verified real-world outcomes rather than speculative or incomplete data. Furthermore, I normalised the raw timestamps into discrete calendar days. This standardisation allows us to capture a stable daily consensus rather than isolated moments in time, providing a consistent baseline for comparing our five aggregation methodologies.

To maintain mathematical stability across these methods, I addressed the zero-probability issue inherent in geometric calculations. Since geometric formulas collapse when processing a 0% or 100% forecast, I applied a standard boundary clipping (0.1% to 99.9%). For the geometric mean of odds, I implemented an even more conservative 1% threshold. This technique preserves the critical signal from highly confident forecasters while preventing any single extreme certainty from invalidating the group's collective wisdom. This ensures the resulting aggregates are robust, extremised where appropriate, and mathematically sound for executive-level decision-making.

## Step 3

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To determine which aggregation method provided the most reliable steer, the Brier score was measured for each approach.

| Method            | Mean Brier Score |
|-------------------|------------------|
| Geom Mean of Odds | 0.110            |
| Trimmed Mean      | 0.112            |
| Raw Mean          | 0.113            |
| Geometric Mean    | 0.114            |
| Median            | 0.114            |

*Note: The Brier score measures the error rate. Lower scores indicate higher accuracy.*

The geometric mean of odds (GMO) was the most accurate method, outperforming all linear aggregates. The GMO is an extremising technique designed to push all groups' consensus closer to 0% or 100%, based on the assumption that forecasters systematically under-report their true confidence. This superior performance suggests that RCTA forecasters did show this hedging behaviour, and their collective knowledge was more certain than their average probability implied. The linear methods (raw mean, trimmed mean and median) are purely descriptive and report what the crowd said well, but in doing so, they preserve the dampened signal. The standard geometric method performed poorly because it extremises the probabilities directly rather than in log-odds space. This meant it collapsed toward 0% for questions where the crowd was largely uncertain. I validated the GMO's superiority using the Diebold-Mariano test against the nearest competitor (the trimmed mean,  $DM = -12.3$ ,  $p < 0.001$ ), confirming the result is statistically robust.

#### Step 4

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The geometric mean of odds was identified as the strongest baseline for this dataset. However, even this method has a structural weakness in that it treats every forecaster's opinion as equally valuable. Forecasting skill varies considerably in practice, with some individuals being systematically more accurate than others across questions. A more powerful aggregate

should amplify the voices of proven forecasters and discount those with poor track records.

To achieve this, I developed a skill-weighted geometric mean of odds. For each forecaster, I computed their historical mean brier score across all resolved questions using their most recent forecast. I then assigned each forecaster a weight of  $1 / (\text{historical\_brier} + 0.05)$ , making better forecasters contribute more strongly to the log-odds average. The 0.05 is a deliberate cap on how much a single forecaster's weight can exceed the average, to prevent a small number of star performers from dominating the crowd. I also required a minimum track record, with forecasters who had predicted on fewer than five resolved questions, not given a personalised weight due to little evidence on their accuracy.

Key adaptations of this model:

- Prior accuracy: Forecaster weights are derived entirely from performance on previously resolved questions.
- Regularisation: The epsilon floor prevents extreme weight disparities that would destabilise the aggregate when a highly-weighted forecaster happens to be wrong.
- Robust fallback: Forecasters with no qualifying history are not excluded from the aggregate, but they contribute at average weight. This preserves crowd breadth on questions with few experienced participants.

| Method                                | Mean Brier Score |
|---------------------------------------|------------------|
| Skill-Weighted Geometric Mean of Odds | 0.1098           |
| Geom Mean of Odds                     | 0.1104           |
| Trimmed Mean                          | 0.1123           |
| Raw Mean                              | 0.1125           |
| Geometric Mean                        | 0.1136           |
| Median                                | 0.1138           |

*Note: Lower scores indicate higher accuracy. Scores are provided to four decimal places to highlight the precision gain achieved by the improved model.*

This approach outperforms the five standard benchmarks because it introduces genuinely new information, individual forecaster quality, rather than simply transforming the same aggregate signal. The standard GMO implicitly assumes the crowd is homogeneous, whilst the skill-weighted version recognises that within any crowd some voices have a better track record. The improvement is modest in absolute terms (0.0006 Brier points) but meaningful as it represents a consistent, systematic gain over the best available benchmark.